

Innovating Biomedical Devices Student Competition & Showcase 2026

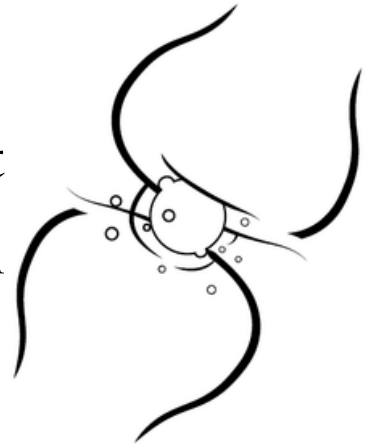
Non-invasive and Adaptive Brain-Computer Interface Based on Electroencephalographic Signals from the Primary Motor Cortex



NeuroSynergy Team
Exploring Neural Frontiers

Presented and Authored by Giuliano Agostini

SyNexus Project
Innovating Neural Connection



Politecnico
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INTRODUCTION

The core concepts and fundamental principles

Versatility

The quality of being able to adapt or be adapted to many different functions, activities, or situations



A system designed to adapt to each patient's individual characteristics and needs

Practicability

The quality of being feasible or capable of being effectively carried out or put into practice



Finding a solution that is both effective and realistically achievable, for a complex problem

Well-being

The state of being comfortable, healthy, or happy, and a state of physical, mental, and social welfare



The system revolves around the patient, ensuring physical and mental well-being through close collaboration with the experimenter



THE CLINICAL CHALLENGE

Neuropathology, Neurotrauma, and the Neuroanatomical Basis of
Motor Control

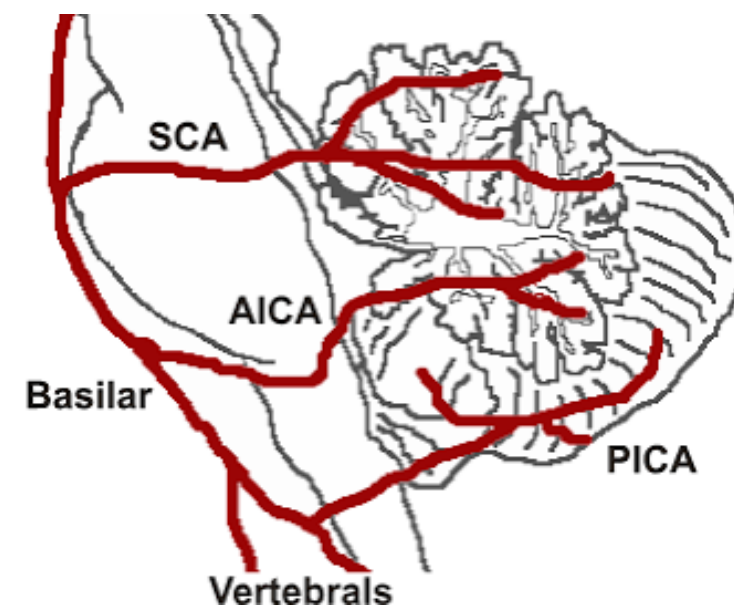
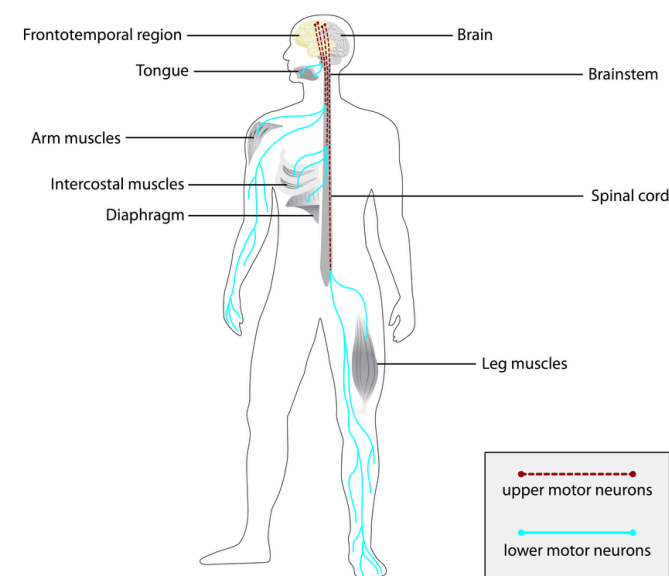


THE ROOTS OF THE PROBLEM

When the Nervous System Fails to Communicate and Neural Control Breaks Down

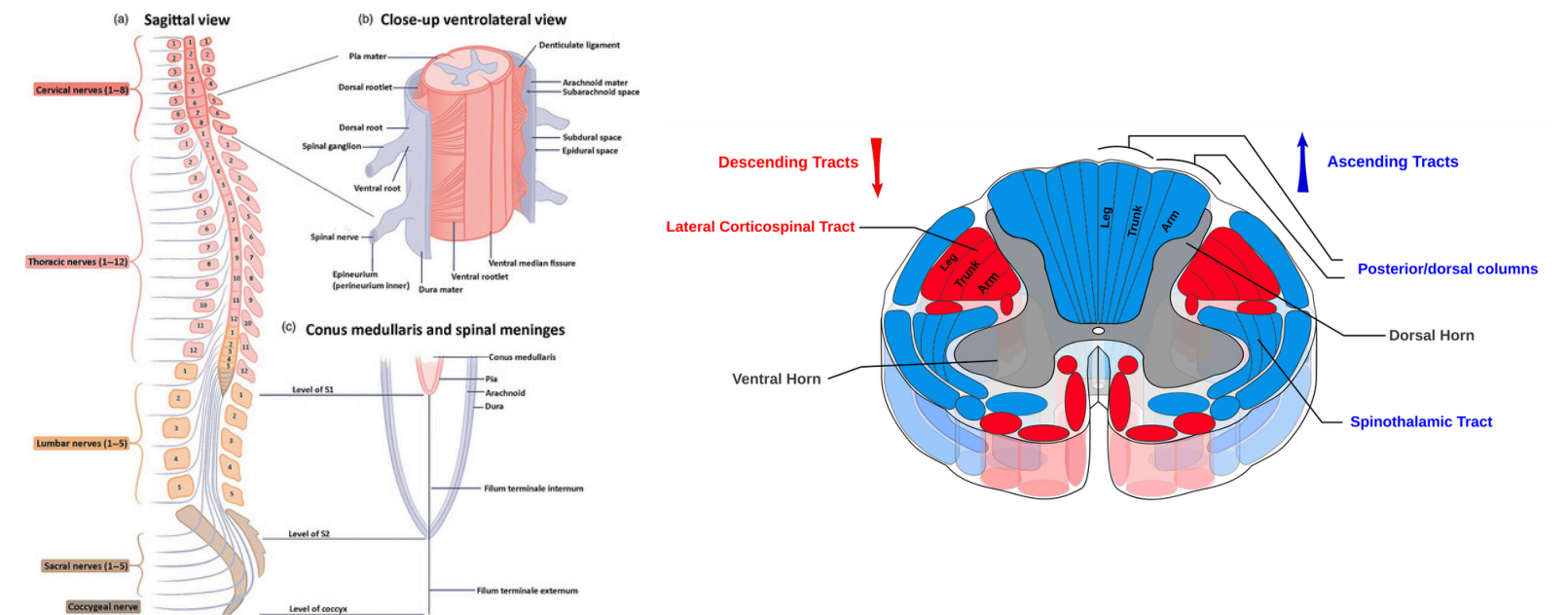
NEUROPATHOLOGY

Pathologies such as Amyotrophic Lateral Sclerosis and Locked-in Syndrome bring patients to a state of paralysis



NEUROTRAUMA

Trauma due to accidents can damage the spinal cord and stop the communication between the Central Nervous System and the muscles



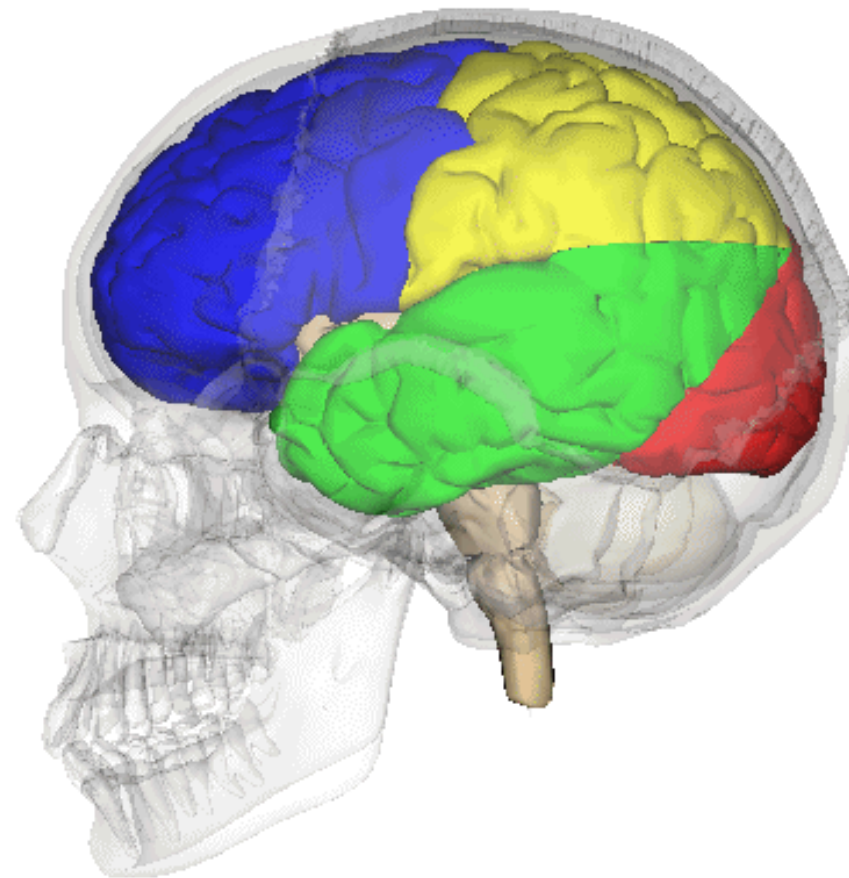
THE CENTRAL NERVOUS SYSTEM

Hemispheres, Lobes, and Functionality of the Encephalon

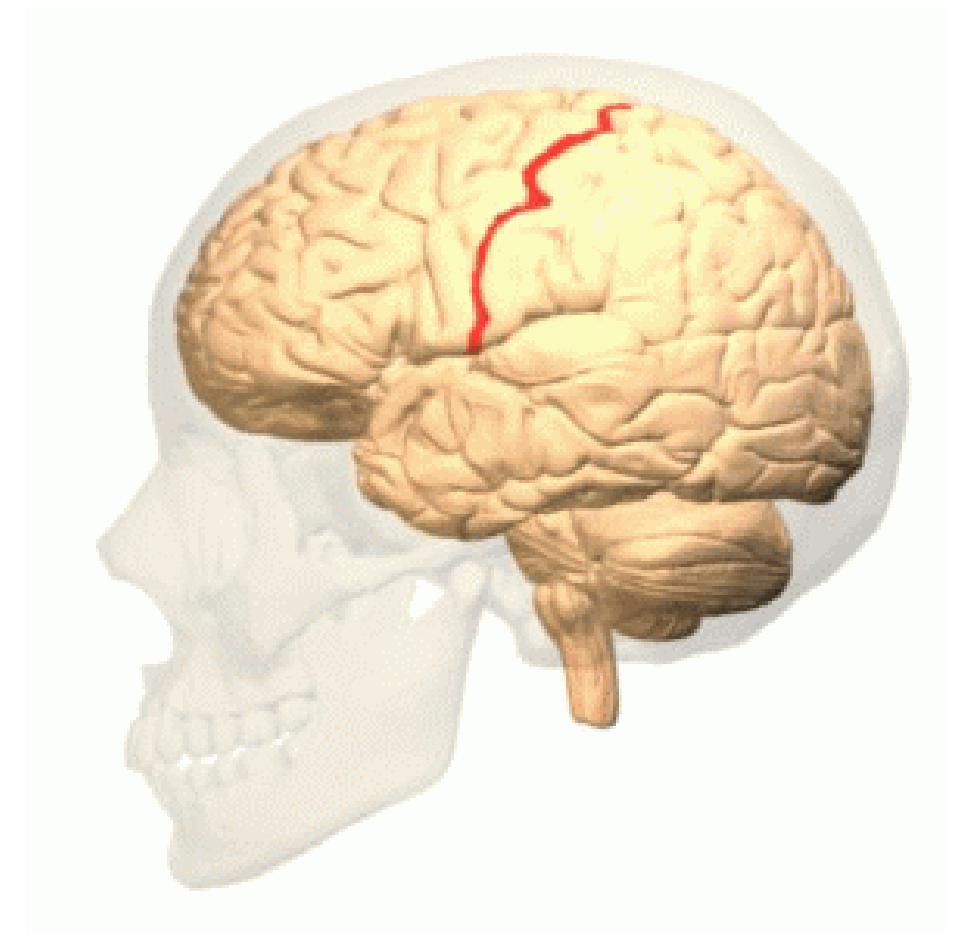
Cerebral Hemispheres
Functional Division



Cerebral Lobes
Functional Localisation



The Central Sulcus
Functional Specialisation

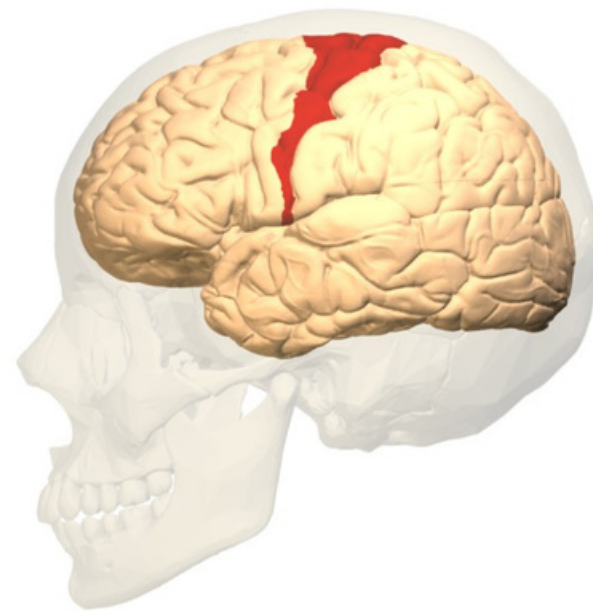


THE PRIMARY MOTOR CORTEX

Anatomical and Functional Basis of Motor Control

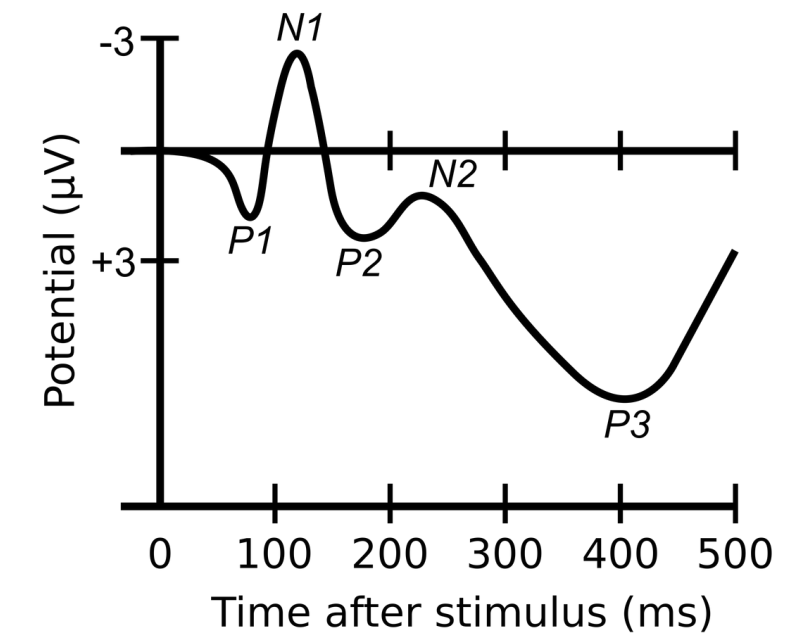
Anatomical Structure

- Between the Frontal and the Parietal Lobe
- Communication through Motor Units and Pyramidal System



Functional Characteristics

- Presence of Event-Related Potentials (ERPs)
- Correlation with expression of physical and imagined movement



HARDWARE AND SIGNAL ACQUISITION

Wearable Design Principles, Electrode Materials, and the Electronic Chain
Behind Signal Acquisition

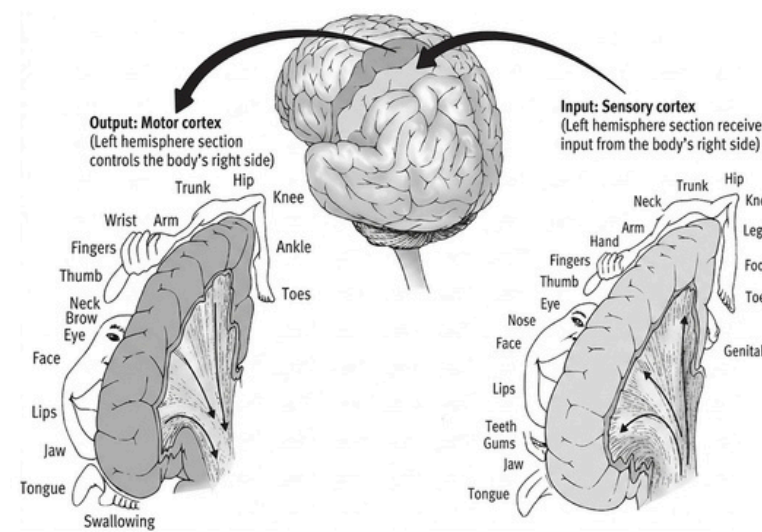
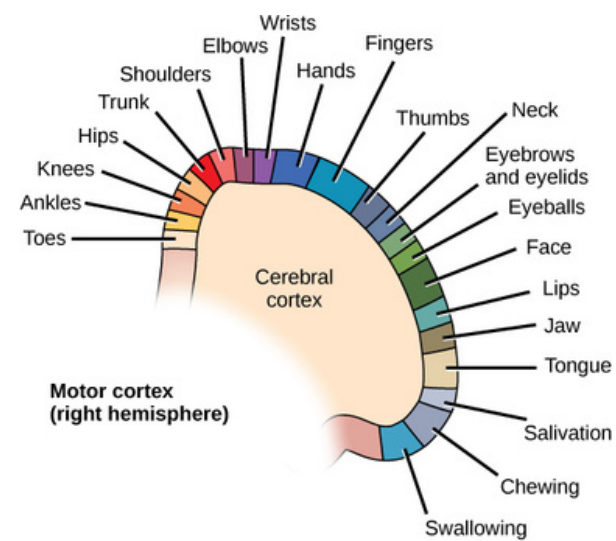
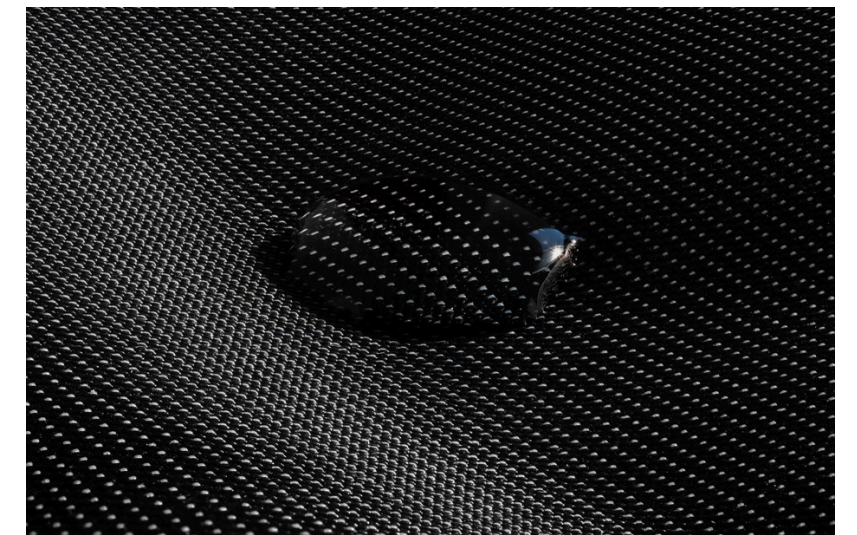
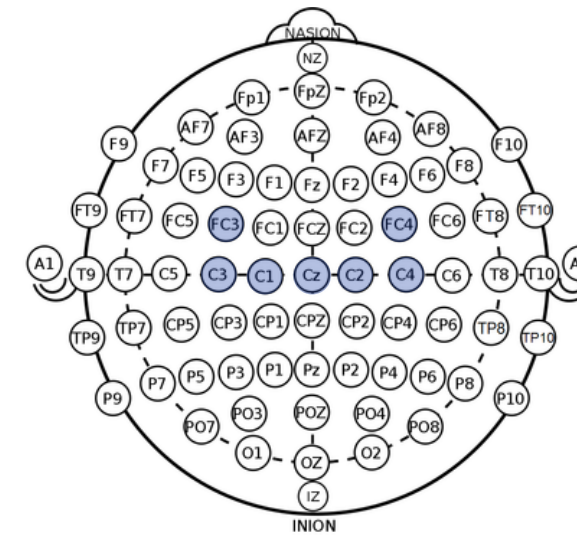


HEADSET AND ELECTRODE PLACEMENT

Wearable Design Principles and Localized Electrode Positioning

The aim is to provide an headset that guarantees:

- Patient's comfort by avoiding an invasive EEG Cap
- Modularity by choosing an adaptable elastic structure
- Biocompatibility, by choosing hypoallergenic materials



The electrode placement focuses on:

- An adapted structure based on the 10-20 system
- Acquisition of the hand and arm representation
- Cognitive events linked to motor impairment



ELECTRODES AND COMMUNICATION

Electrode Materials and Signal Transmission Pathways

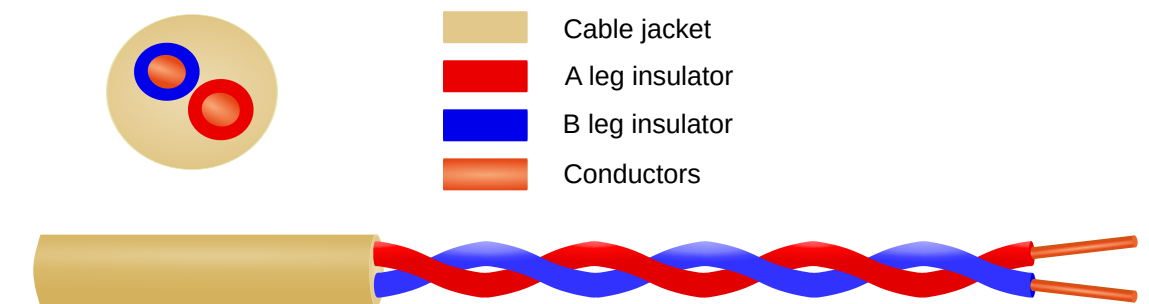


The applied parts have the following characteristics:

- Ag/AgCl surface, optimal for biosignal acquisition
- Thermoplastic elastomer structure for long-term use
- Dry acquisition, avoiding the use of conductive paste

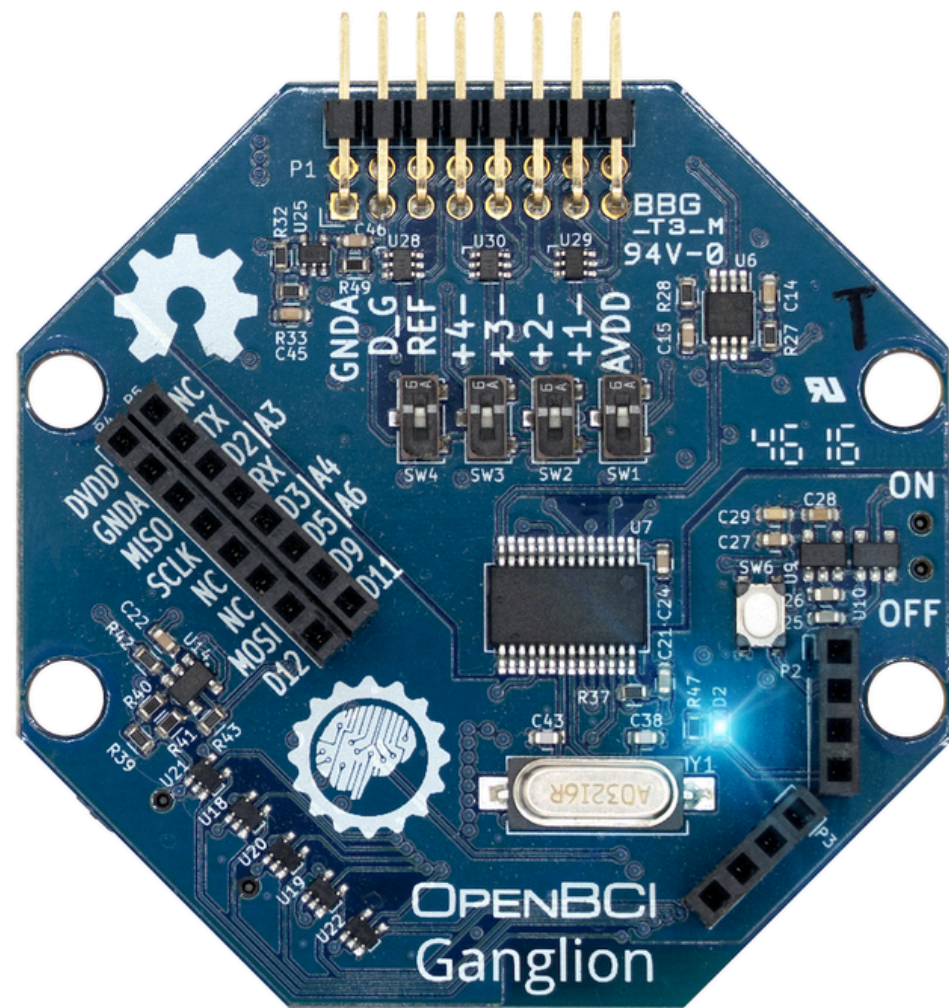
The transmission of the signal involves:

- Thin and light structure to minimise invasiveness
- Coated external insulation
- Twisting technique, to minimise electromagnetic noise



THE PROCESSING BOARD

The Electrode Wiring Scheme, On-Body Placement Configurations, and Processing Board's Circuitry,

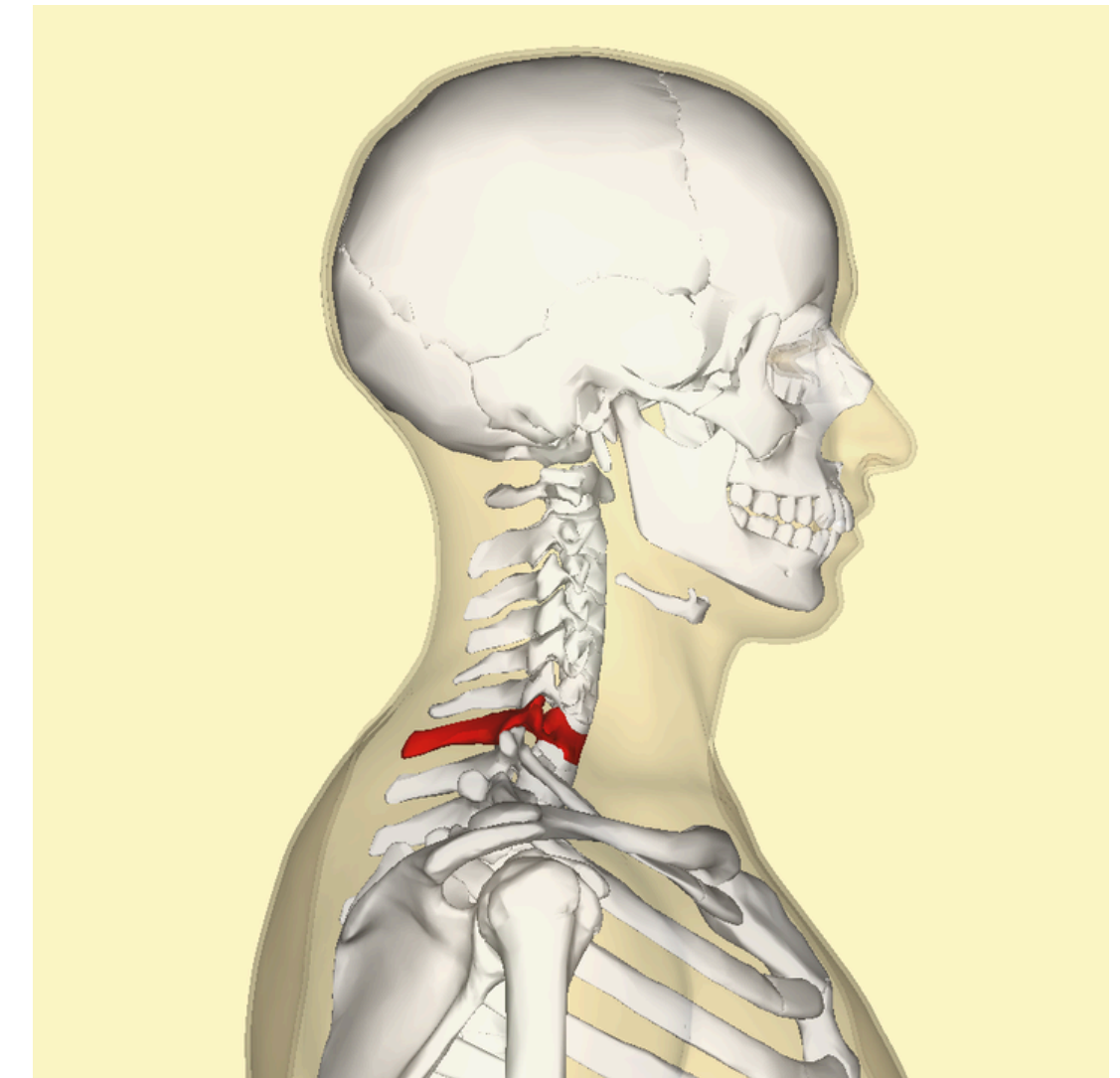


The core of the hardware system is the OpenBCI
Ganglion Board

Connected to the electrodes through a 16-pin
header

Equipped with an AD8237 instrumentation
amplifier

The board is battery-powered and positionable at
the level of the C7 vertebra



SOFTWARE AND PROCESSING ARCHITECTURE

Turning Raw Neural Signals into Real-Time Commands through an
Integrated Software Pipeline



SOFTWARE-HARDWARE INTERACTION

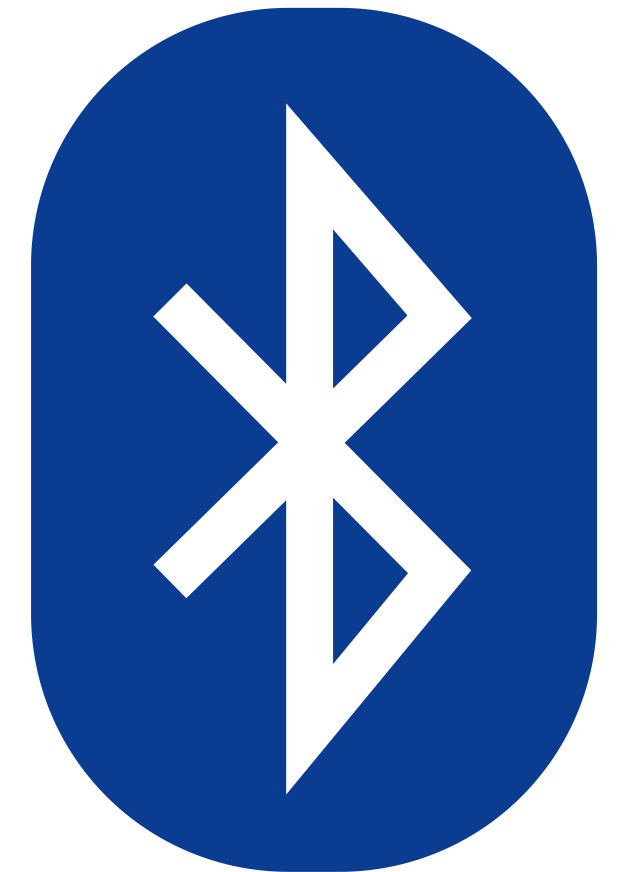
Establishing a Stable Digital Pathway from Raw Signal Acquisition to Reliable Command Transduction

OBJECTIVE 1

Implement open-source platforms and prioritise collaboration between experimenter and patient

OBJECTIVE 2

A stable and wireless communication through USB connector and the native bluetooth feature of the Board



INVOLVED DEVELOPMENT PROGRAMS

An Integrated Pipeline for Signal Visualization, ERP Analysis, and Real-Time Command Transduction

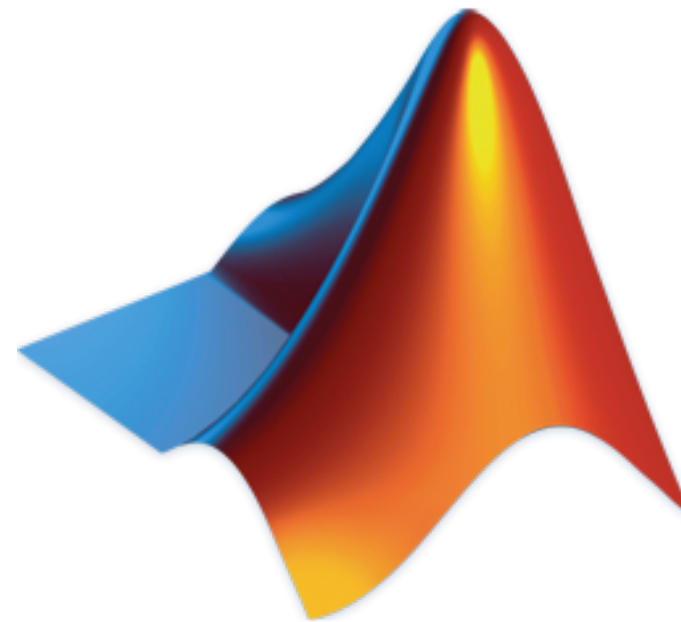
OPENBCI GUI

Establishing the first connection with the board
Monitoring and adjusting the positioning of the electrodes
Verifying the EEG signal quality



MATLAB

Cleaning the raw EEG signal through EEGLAB
Identifying and studying ERP phenomena with ERPLAB
Creating a training support with App Designer



BRAINFLOW

Definition of the transduction method for the analysed signal
Establishment of the command-generation criteria
Creation of the standalone application for cursor control



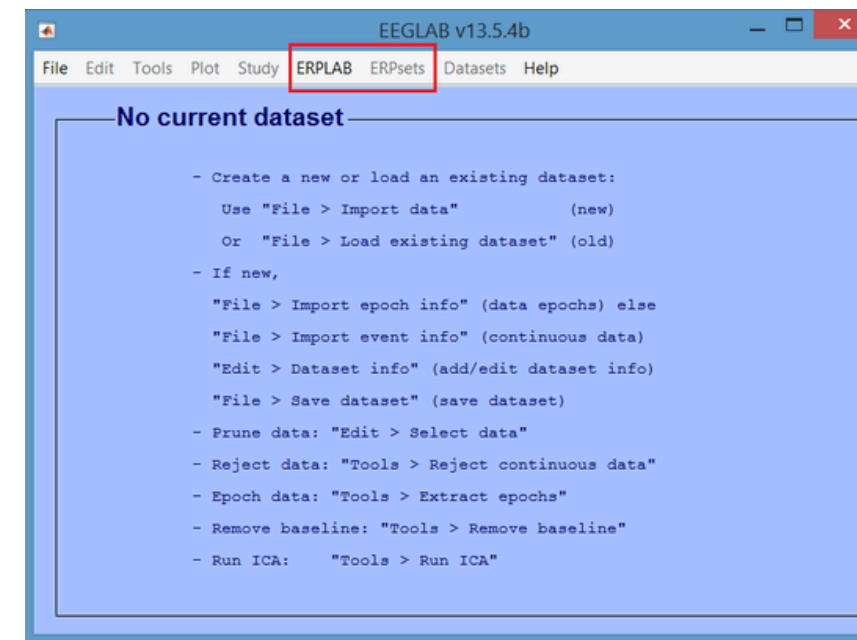
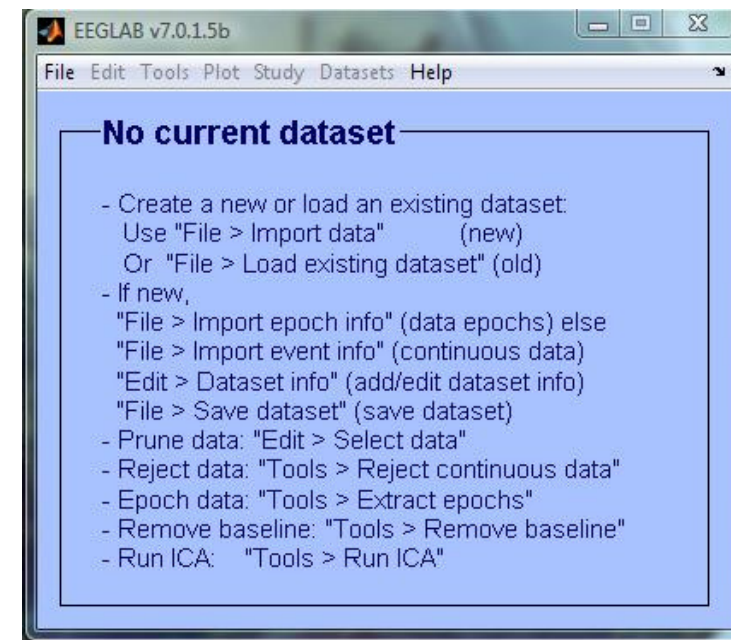
MULTI-SOFTWARE ENVIRONEMENT

Recording, Analysing, Training, and Controlling in a Unified Workflow

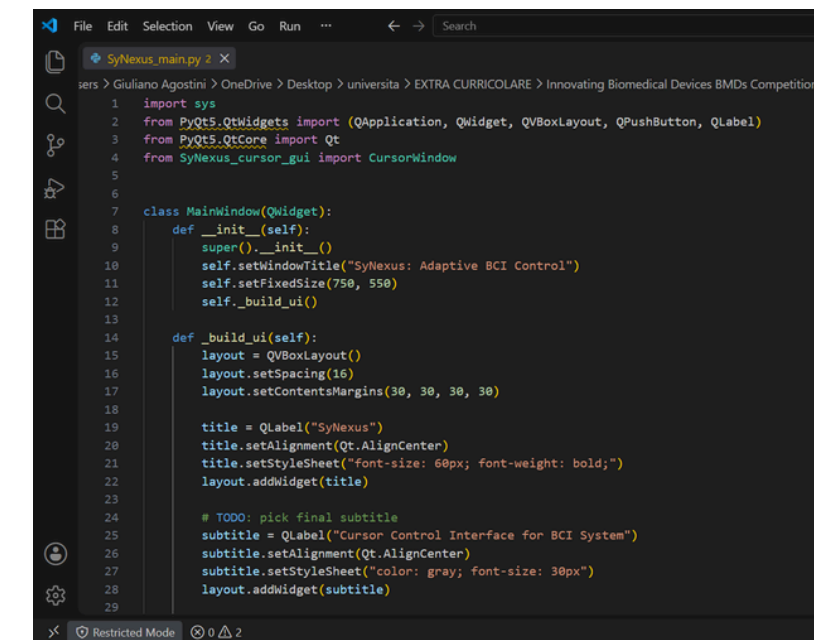
OpenBCI GUI
Signal Recording



MATLAB
Signal Processing and Training



Brainflow
Device Control



METHODOLOGY AND EXPERIMENTATION

Designing and Validating the Experimental Approach Behind the BCI System



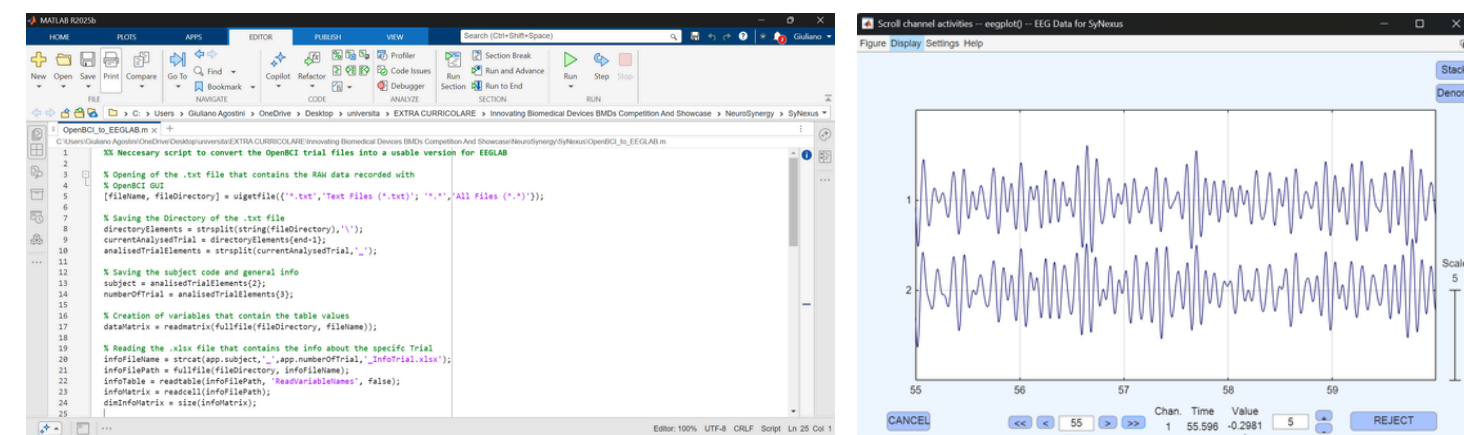
THE OPERATIONAL PIPELINE

From Patient Assessment to Real-Time Cursor Control

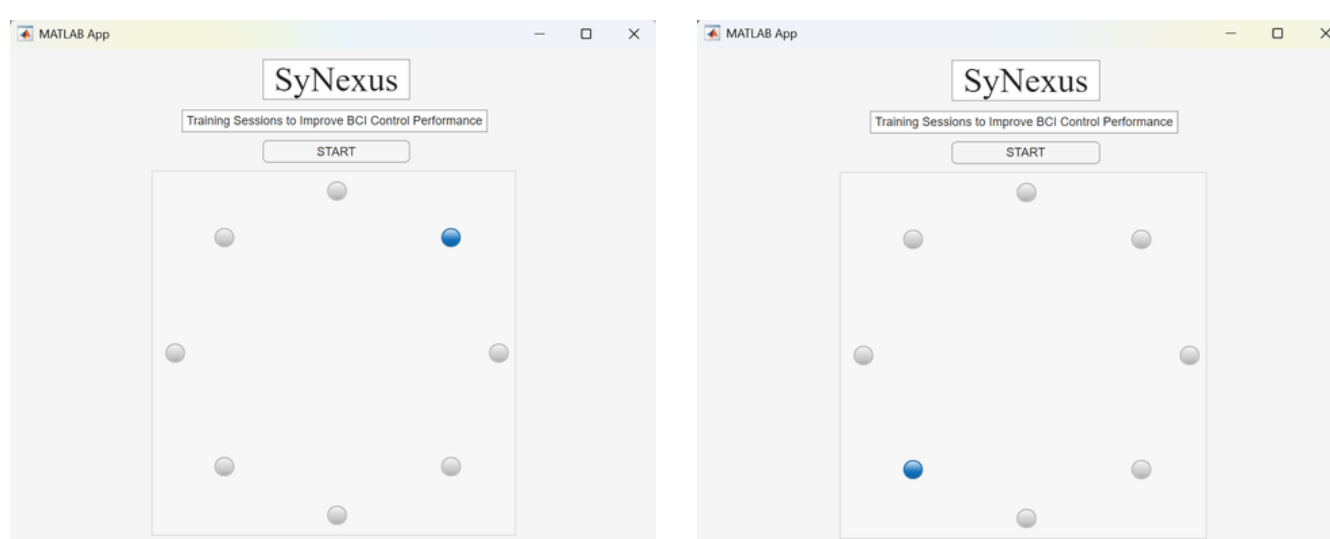
1 Patient Intake and EEG Acquisition



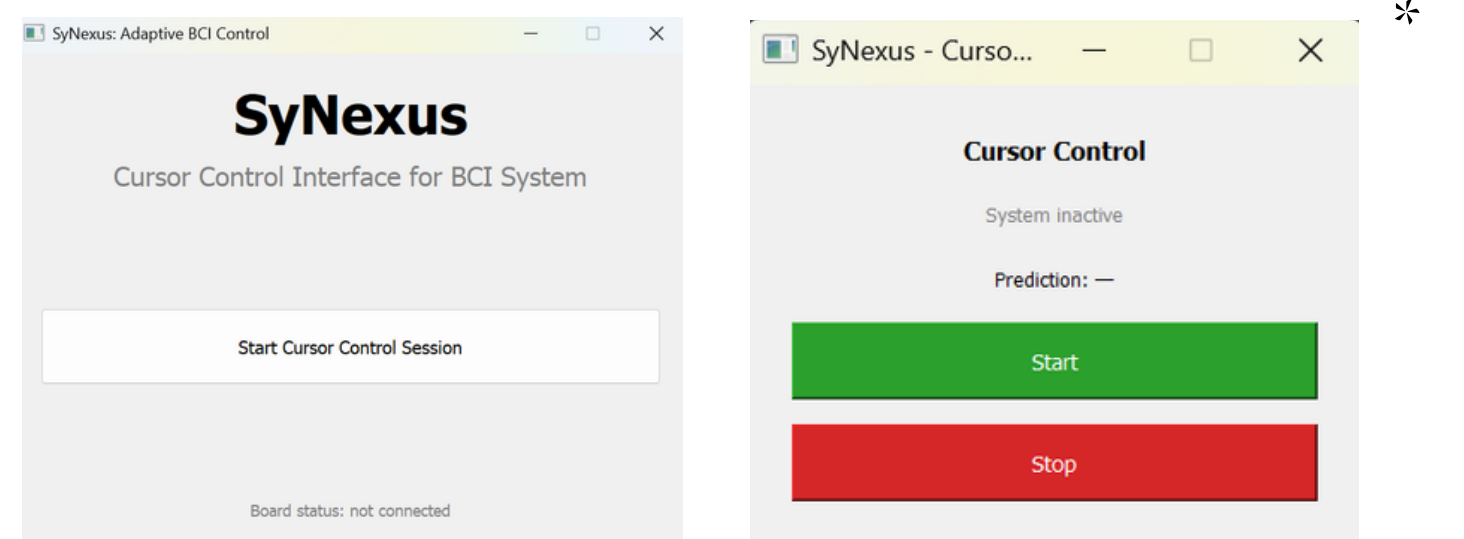
2 EEGLAB and ERPLAB Analysis



3 Training Session



4 Cursor Control



FUTURE DEVELOPMENTS

Extending the System Beyond the Current Prototype, Toward the Next Steps
for Hardware, Software, and Patient Well-Being



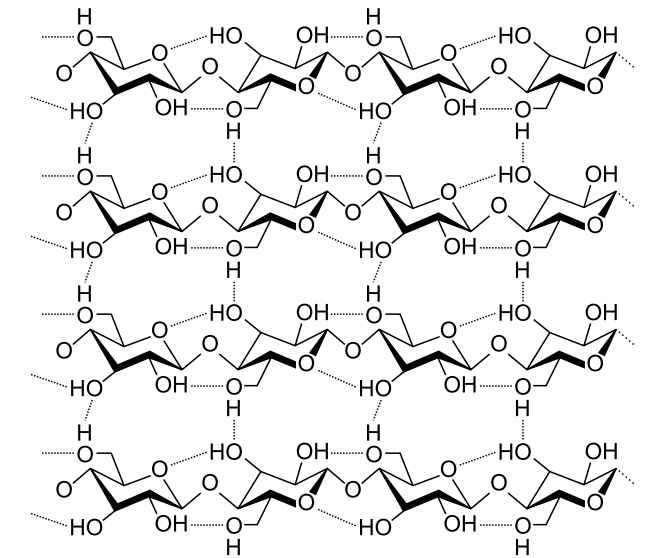
BEYOND THE PROTOTYPE

Hydrogel Electrodes, Wireless Communication, and an AI-Assisted, Interoperable Application

HARDWARE UPGRADES

Moving to innovative materials such as biopolymers and microarray electrodes, to improve signal acquisition

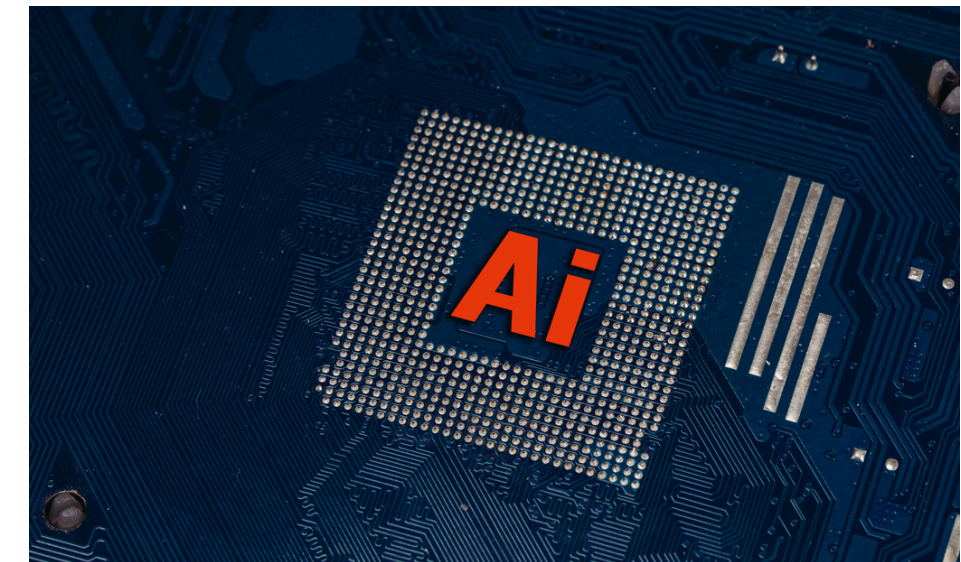
Improving comfort with biocompatible electrodes, capable of adapting to each patient's individual anatomy



SOFTWARE UPGRADES

Expanding digital compatibility to portable devices such as smartphones and tablets, for everyday use

Creating an environment where clinicians can actively monitor the patient, and where AI can be implemented to support them



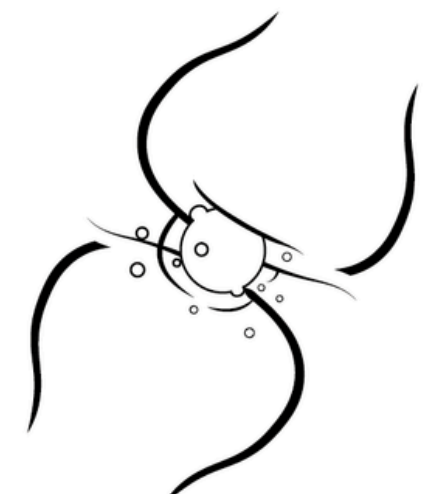
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Thank you for your attention

The SyNexus project was born from the will to delve into Brain-Computer Interfaces, demonstrating how neurology, electronics, materials science, signal analysis, and informatics can converge under a single purpose



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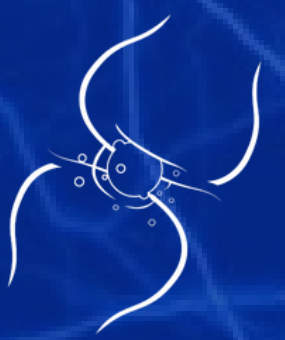


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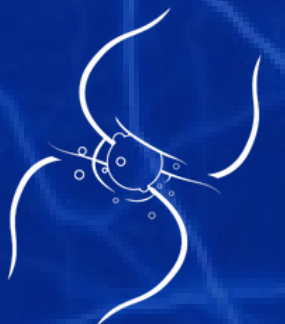


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- [F1] Diagram of a human highlighting the areas affected by amyotrophic lateral sclerosis (ALS) including the upper motor neurons, lower motor neurons, and frontotemporal region of the brain, along with the muscles affected as a result. Attribution: PaulWicks, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons
- [F2] Diagram of cerebellum blood supply. Attribution: Semiconscious, Public domain, via Wikimedia Commons
- [F3] Spinal cord segments, vertebral view, and conus medullaris. Attribution: Sheryl Tan, Faull L, Curtis A, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons
- [F4] Cerebral hemispheres. Attribution: Polygon data were generated by Database Center for Life Science (DBCLS), CC BY-SA 2.1 JP <<https://creativecommons.org/licenses/by-sa/2.1/jp/deed.en>>, via Wikimedia Commons. Modification: Screenshot taken from the original animation.
- [F5] four lobes. frontal lobe(blue), parietal lobe(yellow), temporal lobe(green) and occipital lobe(red) of left hemisphere. and also corpus callosum, diencephalon and brainstem are drawn. Images are from Anatomography maintained by Life Science Databases(LSDB). Attribution: Images are generated by Life Science Databases(LSDB)., CC BY-SA 2.1 JP <<https://creativecommons.org/licenses/by-sa/2.1/jp/deed.en>>, via Wikimedia Commons. Modification: Screenshot taken from the original animation.
- [F6] Central DBCLS[2]., CC BY-SA 2.1 JP <<https://creativecommons.org/licenses/by-sa/2.1/jp/deed.en>>, via Wikimedia Commons. Modification: Screenshot taken from the original animation.
- [F7] Brodmann areas 4. Primary motor cortex. Attribution: 3D brain data is from Anatomography., CC BY-SA 2.1 JP <<https://creativecommons.org/licenses/by-sa/2.1/jp/deed.en>>, via Wikimedia Commons
- [F8] Brodmann areas 4. Primary motor cortex. Attribution: 3D brain data is from Anatomography., CC BY-SA 2.1 JP <<https://creativecommons.org/licenses/by-sa/2.1/jp/deed.en>>, via Wikimedia Commons
- [F9] Vectorization and slight modification of Constudevent.gif. Attribution: Original: Choms Vector:Mononomic, CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons.



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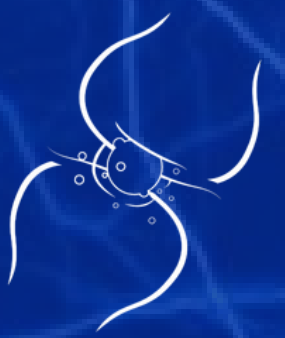


References and Figures (3)

- [F10] International 10-10 system for EEG electrode placement, showing modified combinatorial nomenclature. This is the correct version of the existing 10-20 system where the electrode positions CP8, CP10, FC8 and FC10 have been corrected to TP8, TP10, FT8 and FT10. Attribution: Brylie Christopher Oxley, CCo, via Wikimedia Commons
- [F11] Drop of water on water-resistant textile (100% polyester). Attribution: Petar Milošević, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons
- [F12] Different parts of the motor cortex control different muscle groups. Muscle groups that are neighbors in the body are generally controlled by neighboring regions of the motor cortex as well. For example, the neurons that control finger movement are near the neurons that control hand movement. Attribution: CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons
- [F13] The sensory and motor cortex. Attribution: PsychologyAssociation, CCo, via Wikimedia Commons
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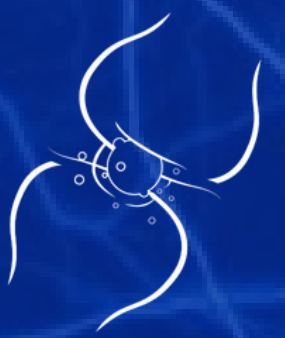


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- [F21] Logo of Bluetooth (IEEE 802.15.1), a bind rune merging the Younger Futhark runes $\mathfrak{H}$ and $\mathfrak{B}$, the initials of Harald Bluetooth. Jnmasek, Public domain, via Wikimedia Commons
- [F22] OpenBCI. OpenBCI Documentation. OpenBCI logo. https://docs.openbci.com/?_gl=1*itg9ub*_gcl_au*ODcoMDIzOTAuMTc4Nzg3NjkoMw..*_ga*MTErODkoNzY4MC4xNzg3ODM2OTQz*_ga_HVMLCoZWWS*czE3ODc5MDY5MTYkbzlkZzEkdDE3ODc5MDg3MTQkaJYwJGwwJGgw
- [F23] First eigenfunction of the L-shaped membrane, resembling (but not identical) to MATLAB's logo trademarked by MathWorks Inc. Attribution: By MathWorks - /usr/local/MATLAB/R2023b/bin/glnxa64/cef_resources, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=142676055>
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References and Figures (5)

- [F31] Schematic image to illustrate the Google Pixel 7a. Simple geometry primitives used to build up shapes. Attribution: JorgenQ, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons
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